

A project report on

CLASSIFICATION OF SPINAL CORD DISORDERS USING INCEPTION-V3

Submitted in partial fulfillment for the award of the degree of

Masters of Technology in Computer Science and Engineering

by

SAYANTANI GUHA (23MCS1007)



VIT[®]

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

November, 2024

A project report on

CLASSIFICATION OF SPINAL CORD DISORDERS USING INCEPTIONV3

Submitted in partial fulfillment for the award of the degree of

Masters of Technology in Computer Science and Engineering

by

SAYANTANI GUHA (23MCS1007)



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

November, 2024



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

DECLARATION

I hereby declare that the thesis entitled “CLASSIFICATION OF SPINAL CORD DISORDERS USING INCEPTION-V3” submitted by Sayantani Guha (23MCS1007), for the award of the degree of Masters of Technology in Computer Science and Engineering, Vellore Institute of Technology, Chennai is a record of bonafide work carried out by me under the supervision of Dr Amutha S.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Chennai

Date:

Signature of the Candidate



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

School of Computer Science and Engineering

CERTIFICATE

This is to certify that the report entitled “CLASSIFICATION OF SPINAL CORD DISORDERS USING INCEPTION-V3” is prepared and submitted by **Sayantani Guha (23MCS1007)** to Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **Masters of Technology in Computer Science and Engineering** is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

Signature of the Guide:

Name: Dr./Prof.

Date:

Signature of the Examiner

Name:

Date:

Signature of the Examiner

Name:

Date:

Approved by the Head of Department,
Masters of Technology in Computer Science and Engineering

Name: **Dr. G Malathi**
Professor

Date:

(Seal of SCOPE)

ABSTRACT

This study investigates the use of deep learning for detecting and classifying spinal disorders, specifically spondylolisthesis and scoliosis, using vertebrae X-ray images. Spondylolisthesis involves the forward slippage of a vertebra, leading to pain and potential neurological issues, while scoliosis is marked by an abnormal lateral curvature of the spine, causing varying levels of physical and functional impairments. The early detection of these conditions is essential for effective management and treatment.

The research employs the InceptionV3 deep learning model to classify X-ray images into three categories: normal, spondylolisthesis, and scoliosis. A dataset sourced from King Abdullah University Hospital and Jordan University of Science and Technology, containing images preprocessed and resized for model compatibility, was used for training and evaluation. The model achieved an impressive classification accuracy of 96%, demonstrating its effectiveness in identifying spinal abnormalities and its potential to enhance clinical workflows by reducing reliance on manual assessments.

This work emphasizes the role of artificial intelligence in advancing medical imaging, showcasing how deep learning can support healthcare professionals in making more accurate and efficient diagnoses. The study not only illustrates the feasibility of AI-driven solutions for spinal diagnostics but also highlights their scalability and applicability in real-world medical scenarios.

Future research aims to improve the model's generalizability by incorporating larger and more diverse datasets and validating its performance across broader populations. Such developments could further enhance its clinical utility and solidify its role in modern medical practice.

ACKNOWLEDGEMENT

It's my pleasure to express with deep sense of gratitude to Dr. Amutha S, Assistant Professor, School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, for her constant guidance, continual encouragement, understanding; more than all, she taught me patience in my endeavor. My association with her is not confined to academics only, but it is a great opportunity on my part of work with an intellectual and expert in the field of Deep Learning.

It is with gratitude that I would like to extend thanks to our honorable Chancellor, Dr. G. Viswanathan, Vice Presidents, Mr. Sankar Viswanathan, Dr. Sekar Viswanathan and Dr. G V Selvam, Executive Director Dr. Sandhya Pentareddy, Assistant Vice-President, Ms. Kadhambari S. Viswanathan, Vice-Chancellor, Dr. V. S. Kanchana Bhaaskaran, Pro-Vice Chancellor Dr. T. Thyagarajan and Additional Registrar, Dr. P.K. Manoharan for providing an exceptional working environment and inspiring all of us during the tenure of the course.

Special mention to Dean, Dr. Ganesan R, Associate Dean Academics, Dr. Parvathi R and Associate Dean Research, Dr. Geetha S, SCOPE, Vellore Institute of Technology, Chennai, for spending their valuable time and efforts in sharing their knowledge and for helping us in every aspect.

In a jubilant mood, I express ingeniously my whole-hearted thanks to Dr. G. Malathi, Head of the Department and Dr. N.M. Elango, Project Coordinator, SCOPE, Vellore Institute of Technology, Chennai, for their valuable support and encouragement to take up and complete the thesis.

My sincere thanks to all the faculties and staffs at Vellore Institute of Technology, Chennai who helped me acquire the requisite knowledge. I would like to thank my parents for their support. It is indeed a pleasure to thank my friends who encouraged me to take up and complete this task.

Place: Chennai

Date: 20/11/2024

SAYANTANI GUHA

CONTENTS

CONTENTS	iii
LIST OF FIGURES	iv
LIST OF TABLES	iv

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION	5
1.2 OVERVIEW OF PROJECT	6
1.3 CHALLENGES PRESENT IN PROJECT	7
1.4 PROJECT STATEMENT	9
1.5 OBJECTIVES AND SCOPE OF THE PROJECT	10
1.6 CONTRIBUTION TO THE PROJECT	14

CHAPTER 2

RELATED STUDY

2.1 LITERATURE SURVEY	19
2.2 BACKGROUND STUDY TABLE	21

CHAPTER 3

PROPOSED ARCHITECHTURE

3.1 DATASET	22
3.2 DATA AUGMENTATION AND DATA PRE-PROCESSING	23
3.3 INCEPTIONV3 ARCHITECTURE	23
3.4 ADVANTAGES AND DISADVANTAGES....	27

CHAPTER 4

PROPOSED METHODOLOGY

4.1 PSEUDOCODE.....	30
---------------------	----

CHAPTER 5

RESULTS AND DISCUSSION

5.1 PERFORMANCE METRICS EXPLANATION.....	32
5.2 RESULTS.....	34

CHAPTER 6

CONCLUSION

6.1 CONCLUSION.....	37
---------------------	----

LIST OF FIGURES

3.1 INCEPTIONV3 ARCHITECTURE

5.2.3CONFUSION MATRIX

5.2.1 ACCURACY Vs VAL_ACCURACY
PER EPOCHS

5.2.2 ACCURACY & VAL_LOSS PER
EPOCH

LIST OF TABLES

2.2 LITERATURE SURVEY	21
5.1 PERFORMANCE METRICS EXPLANATION	32

Chapter 1

Introduction

1.1 INTRODUCTION

This report delves into the use of deep learning for diagnosing spinal cord disorders, specifically spondylolisthesis and scoliosis, using vertebrae X-ray images. Spondylolisthesis, characterized by the forward displacement of one vertebra over another, can lead to pain, nerve compression, and mobility issues if left untreated. In contrast, scoliosis, involving an abnormal lateral curvature of the spine, ranges in severity from mild deformities to cases causing posture imbalance, chronic pain, and respiratory complications. Early and accurate detection of these conditions is crucial for effective treatment and management. To address this, we employed the Inception V3 model, a robust deep learning architecture known for its superior feature extraction capabilities.

The dataset for this study was collected from King Abdullah University Hospital and Jordan University of Science and Technology. It comprised X-ray images categorized into three groups: normal spines, spondylolisthesis, and scoliosis cases. Preprocessing steps included resizing the images to ensure compatibility with the Inception V3 model, which was then trained and evaluated on these images. The model achieved an impressive accuracy of 96%, underscoring its potential to automate the diagnostic process, minimize reliance on manual assessment, and improve clinical efficiency.

Spondylolisthesis arises from various causes, such as congenital defects, trauma, degenerative changes, or repetitive stress fractures. Symptoms range from mild lower back pain to severe nerve compression, causing radiating pain, numbness, and muscle weakness. Imaging techniques, including X-rays, CT scans, and MRI, aid in diagnosing and quantifying vertebral slippage. Treatment varies based on severity, from physical therapy and bracing to surgical interventions. Scoliosis, classified as idiopathic, congenital, or neuromuscular, involves abnormal spinal curvatures that may cause deformities, uneven shoulders, and even respiratory issues in severe cases. Radiographic studies confirm the diagnosis, while treatments range from observation and bracing to surgical correction.

The Inception V3 model, developed by Google, is particularly effective for image classification tasks. It uses advanced concepts such as Inception modules and factorized convolutions to learn features at different scales and reduce computational overhead. For this study, the dataset was split into training, validation, and testing sets to optimize model performance. The training data included 984 normal spine images, 488 spondylolisthesis images, and 1,935 scoliosis images. Validation and testing sets consisted of smaller subsets.

Deep learning models like Inception V3 offer transformative potential in medical imaging by enhancing diagnostic accuracy and streamlining clinical workflows. This study demonstrates the efficacy of AI in detecting spinal abnormalities, paving the way for its integration into healthcare systems. Future work will focus on expanding the dataset, validating the model across diverse populations, and exploring additional AI architectures to further improve diagnostic reliability and generalizability.

1.2 OVERVIEW OF PROJECT

This project explores the application of deep learning techniques, specifically using the Inception V3 model, to classify spinal cord disorders such as spondylolisthesis and scoliosis from vertebrae

X-ray images. Spondylolisthesis, characterized by the forward slippage of one vertebra over another, and scoliosis, involving an abnormal lateral curvature of the spine, are two conditions that can significantly impact a patient's quality of life. Timely diagnosis and intervention are crucial to prevent progression and manage symptoms effectively. The project aims to automate the diagnostic process, offering a faster, more accurate, and objective alternative to traditional manual methods.

The study utilizes a dataset sourced from King Abdullah University Hospital and Jordan University of Science and Technology, consisting of X-ray images categorized as normal, spondylolisthesis, and scoliosis cases. The dataset underwent preprocessing, including resizing and normalization, to ensure compatibility with the Inception V3 model. The architecture of Inception V3, with its advanced features like Inception modules, factorized convolutions, and auxiliary classifiers, allows it to capture intricate patterns in the images, making it ideal for medical imaging applications. By training, validating, and testing the model on this dataset, the study achieved an impressive accuracy of 96%, demonstrating its effectiveness in classifying spinal disorders.

The results highlight the potential of AI to transform healthcare by reducing diagnostic delays and minimizing human error. The project's confusion matrix and evaluation metrics, such as precision, recall, and F1 score, validate the model's performance and reliability. These findings underscore the utility of integrating AI tools into clinical workflows, enabling healthcare professionals to make informed decisions quickly and efficiently. Future work will focus on expanding the dataset to include diverse populations, testing the model's robustness across various imaging conditions, and integrating it into real-time diagnostic systems. This could pave the way for widespread adoption of AI in medical diagnostics, enhancing clinical efficiency and improving patient outcomes. The project serves as a significant step toward leveraging AI for better healthcare delivery.

1.3 CHALLENGES PRESENT IN PROJECT

Below are some of the primary challenges associated with this project:

1.3.1. Dataset Limitations

- The dataset used for this project is relatively small and sourced from a limited number of institutions, which may not accurately reflect the diversity of global populations or the variety of spinal conditions. The small dataset size can lead to overfitting, where the model becomes too tailored to the training data and fails to generalize well to new, unseen data. A more diverse dataset is essential to improve the model's ability to handle variations in image quality, anatomical differences, and subtle deviations in spinal abnormalities.

1.3.2. Class Imbalance

- The dataset exhibits class imbalance, with a significantly higher number of scoliosis images compared to normal and spondylolisthesis cases. This imbalance can negatively impact the model's performance, as deep learning algorithms tend to favor the more frequent class, leading to biased predictions. Addressing this issue requires techniques such as oversampling the minority classes, undersampling the majority class, or using class-weight adjustments during model training to ensure that the model learns to classify all categories equally well.

1.3.3. Image Preprocessing and Quality Variations

- X-ray images often come in varying resolutions and quality, which can present challenges in preprocessing the images for consistent input into the model. Retaining critical diagnostic features while resizing and normalizing the images can be difficult, as poor-quality images may lose important details that are essential for accurate diagnosis. Additionally, differences in X-ray equipment and imaging techniques across institutions can lead to variations in contrast, lighting, and orientation, which further complicates image analysis.

1.3.4. Model Overfitting and Generalization

- Deep learning models, especially complex ones like Inception V3, are prone to overfitting if the training dataset is not sufficiently large or diverse. Overfitting occurs when the model

memorizes the training data rather than learning generalizable features, which harms its performance on unseen data. Careful model tuning, regularization techniques, and the use of validation datasets are necessary to ensure the model generalizes well to real-world clinical settings without relying too heavily on the training data.

1.3.5. Computational Resource Requirements

- Training deep learning models, particularly large ones like Inception V3, requires substantial computational power. This can be a significant challenge, as the high demands for GPU processing, memory, and storage can be prohibitive, especially in environments with limited resources. Efficient optimization techniques and access to powerful hardware or cloud-based computing services are essential for handling the extensive training processes involved in deep learning tasks.

1.4 PROJECT STATEMENT

The project aims to develop an automated system for the classification of spinal disorders, specifically scoliosis and spondylolisthesis, using deep learning techniques applied to X-ray images. Spinal disorders such as spondylolisthesis, characterized by the forward displacement of vertebrae, and scoliosis, defined by abnormal lateral curvature, are conditions that can severely impact a patient's health and quality of life. Early and accurate diagnosis of these conditions is essential for effective treatment, as delays or misdiagnosis can lead to severe complications. Traditionally, diagnosing such conditions requires manual examination by radiologists, which is time-consuming and prone to human error. This project seeks to mitigate these challenges by implementing a deep learning-based system to automatically classify spinal conditions, aiding medical professionals in making faster, more accurate diagnoses.

To achieve this, the project utilizes the Inception V3 model, a state-of-the-art convolutional neural network architecture known for its efficiency in handling image classification tasks. The model is trained on a dataset of vertebral X-ray images sourced

from King Abdullah University Hospital and Jordan University of Science and Technology, consisting of images categorized into normal spines, spondylolisthesis, and scoliosis. The system is designed to process and classify X-ray images with high accuracy, aiming for at least 96% classification performance. By leveraging deep learning, the system is expected to automate the identification of spinal disorders, providing quick and accurate results, which can be directly used by healthcare professionals to inform treatment decisions.

A significant challenge in this project is the class imbalance within the dataset, as there are more scoliosis images compared to normal and spondylolisthesis cases. To address this, techniques such as data augmentation, oversampling of minority classes, and adjustment of class weights are employed during the model training phase. The project also emphasizes the need for high-quality image preprocessing, as the X-ray images can vary in quality due to differences in equipment and imaging techniques. Additionally, efforts are made to ensure the model generalizes well, avoiding overfitting while maintaining high accuracy.

The outcome of this project will provide a powerful tool for clinical applications, enabling more efficient screening and diagnosis of spinal disorders. By automating the classification process, the system reduces the workload on radiologists and allows for quicker decision-making. Furthermore, the model's potential to be integrated into routine clinical workflows signifies its transformative role in medical imaging, enhancing diagnostic accuracy and improving patient outcomes. The future scope of the project includes expanding the dataset, optimizing the model for real-time applications, and testing its performance in broader clinical settings.

1.5 OBJECTIVES AND SCOPE OF THE PROJECT

1.5.1 OBJECTIVES:

1. Develop an Automated System for Spinal Disorder Classification :

The primary objective is to create an automated system that can classify spinal disorders, specifically spondylolisthesis and scoliosis, using deep learning techniques. This system will process vertebral X-ray images and classify them into one of three categories: normal spine, spondylolisthesis, or scoliosis. The goal is to reduce the reliance on manual analysis by radiologists and provide quicker, more accurate diagnostic support.

2. Achieve High Accuracy Using Inception V3 Model :

A key goal is to leverage the Inception V3 deep learning model to achieve high classification accuracy. Inception V3 is known for its efficient architecture and ability to extract meaningful features from images, making it an ideal choice for classifying complex spinal disorders. By fine-tuning the model on a specialized dataset, the project aims for at least 96% accuracy, ensuring that the system can reliably differentiate between normal and abnormal spinal conditions.

3. Address Dataset Imbalance for Balanced Classification :

One challenge in this project is the dataset imbalance, with more scoliosis images compared to spondylolisthesis and normal spine images. To address this, techniques such as data augmentation, oversampling of underrepresented classes, and class-weight adjustments will be used during model training. The objective is to ensure that the model treats all classes equally and does not become biased toward the more frequent category, thus improving the fairness and reliability of the system.

4. Optimize Image Preprocessing for Consistent Input :

The project aims to optimize the image preprocessing pipeline to handle X-ray images of varying quality, resolution, and size. By resizing, normalizing, and augmenting the images, the system ensures that each input is processed uniformly, retaining critical diagnostic features while eliminating noise. The goal is to create a robust preprocessing strategy that helps the model achieve high accuracy across different image qualities and equipment variations.

5. Ensure Model Generalization and Avoid Overfitting :

Overfitting is a common challenge in deep learning models, particularly when training on a limited dataset. To address this, the project will incorporate regularization techniques, such as

dropout, and use validation data to monitor the model's generalization ability. The objective is to ensure that the model performs well not just on the training set but also on unseen data, making it suitable for real-world clinical applications where new data will be encountered regularly.

6. Integrate the Model into Clinical Practice for Efficient Diagnosis

A long-term objective of this project is to integrate the trained model into clinical workflows, enabling medical professionals to quickly and accurately diagnose spinal disorders from X-ray images. By providing a tool that can automatically classify conditions, the system will assist radiologists and orthopedists in making faster decisions, reducing diagnostic delays, and improving patient outcomes.

7. Expand the Dataset and Model for Real-Time Applications

The project also aims to expand the dataset with more diverse and higher-quality images to improve the model's generalizability and robustness. Additionally, the objective is to explore ways to optimize the model for real-time applications, allowing for faster processing of X-ray images and integration with embedded medical devices for on-the-spot diagnosis during clinical consultations. This could revolutionize how spinal disorders are diagnosed and treated in healthcare settings.

Each of these objectives addresses a critical aspect of the project, from developing a powerful and accurate model to ensuring its practicality and reliability in real-world healthcare environments.

1.5.2 SCOPE:

1. Automated Classification of Spinal Disorders

The project focuses on automating the diagnosis of spinal disorders, specifically spondylolisthesis and scoliosis, through the use of deep learning. By leveraging the capabilities of the Inception V3 model, the system will classify vertebral X-ray images into normal, spondylolisthesis, or scoliosis categories. This automation minimizes manual dependency and paves the way for faster, more reliable diagnostic outcomes.

2. Enhancing Clinical Diagnostic Processes

The scope includes integrating the developed model into clinical workflows to assist radiologists and orthopaedists. By providing accurate and rapid classifications, the model aims to reduce diagnostic delays, streamline treatment planning, and improve patient outcomes. Its deployment in real-world

healthcare systems will demonstrate the transformative role of artificial intelligence in medical diagnostics.

3. Dataset Utilization and Expansion

The project makes use of a curated dataset of vertebral Xray images from King Abdullah University Hospital and Jordan University of Science and Technology. This includes images of varying quality, sizes, and classes. The scope also extends to exploring ways to enhance the dataset by incorporating more diverse images, ensuring the model's adaptability to different patient demographics, and improving generalizability.

4. Model Optimization for Accuracy and Efficiency

Within the project's scope is the optimization of the Inception V3 model for high accuracy (targeting 96%) in classifying spinal conditions. This includes finetuning the model through hyperparameter adjustments, addressing data imbalance using augmentation and sampling techniques, and incorporating regularization strategies to avoid overfitting.

5. Preprocessing and Image Handling

The project covers designing a robust image preprocessing pipeline that normalizes, resizes, and augments Xray images for optimal input to the model. This ensures compatibility across various imaging equipment and quality levels, contributing to consistent performance in diverse clinical environments.

6. Validation and Real-World Deployment

A significant part of the project scope involves testing the model's generalization capabilities on unseen data to ensure reliability. The goal is to create a system that can be seamlessly integrated into medical devices or healthcare platforms, enabling real time analysis of Xray images in clinics, hospitals, and remote healthcare centers.

7. Expanding AI's Role in Healthcare

The broader scope of the project lies in demonstrating how artificial intelligence can revolutionize medical imaging. By setting a benchmark with this system, the project lays the groundwork for developing similar models for other medical imaging tasks, such as detecting fractures, tumors, or other

musculoskeletal disorders, thereby contributing to the advancement of AI driven healthcare solutions.

1.6 CONTRIBUTION TO THE PROJECT

1.6.1. Automated Detection of Spinal Disorders

- The project introduces a deep learning-based model for the automated classification of spinal conditions like spondylolisthesis and scoliosis, reducing reliance on manual interpretation of X-ray images.

This project develops an Inception V3-based model capable of accurately identifying and classifying spinal disorders from X-ray images. By automating this process, the model enhances efficiency, minimizing the time and effort required by radiologists for manual assessments. This contributes to improved patient care, particularly in settings where specialists may be limited, by delivering fast and reliable diagnostic outputs.

1.6.2. High Diagnostic Accuracy

- The model achieves an impressive accuracy of 96%, showcasing its potential to outperform traditional diagnostic methods.

The implementation of Inception V3 and the careful preprocessing of input images ensure that the model delivers exceptional classification accuracy. This high accuracy makes it a reliable tool for clinical application, reducing the risks of misdiagnosis. Furthermore, the project highlights the capability of deep learning models to detect subtle abnormalities that might be overlooked in manual diagnoses.

1.6.3. Integration of Advanced Deep Learning Techniques

- The project effectively demonstrates the application of the Inception V3 architecture, showcasing its strengths in feature extraction and classification of medical images.

By leveraging the advanced features of Inception V3, such as inception modules and factorized convolutions, the project demonstrates the adaptability of state-of-the-art deep learning

models in medical applications. This contribution provides a blueprint for future researchers aiming to use similar architectures

for diagnosing other conditions, expanding the utility of such techniques in healthcare.

1.6.4. Improved Clinical Efficiency

- By reducing diagnostic delays and offering consistent results, the project improves the workflow in clinical settings.

The automation introduced by this project contributes to smoother clinical operations by alleviating the workload of medical professionals. Faster and accurate classifications allow clinicians to focus on treatment strategies, improving patient outcomes. This is especially valuable in resource-constrained environments where the availability of specialized radiologists is limited.

1.6.5. Demonstration of AI's Potential in Healthcare

- The project exemplifies the transformative role of AI in medical diagnostics, paving the way for its integration into real-world clinical workflows.

By achieving high performance and practical applicability, this project serves as evidence of how AI can revolutionize diagnostics. It inspires confidence in adopting AI tools in routine healthcare practices, encouraging further investment and research in this area. It also sets a precedent for exploring AI's applications in other medical domains, broadening its impact on healthcare innovation.

1.7. REAL-TIME SYSTEM OPTIMIZATION:

To adapt and optimize the classification model for real-time usage, several key aspects must be addressed, ensuring that the system performs effectively in clinical or operational environments. Below are the optimization steps and their descriptions:

1.7.1 Deployment on Edge or Cloud-Based Platforms

- Optimization: Deploy the trained Inception V3 model on edge devices (like NVIDIA Jetson Nano or Raspberry Pi) or cloud platforms (like AWS, Azure, or Google Cloud) for real-time analysis.

- Description: Edge deployment reduces latency as computations occur locally, making it suitable for clinics with limited internet access. Conversely, cloud deployment ensures scalability, enabling remote access for diagnostics from various locations.

1.7.2. Lightweight Model Implementation

- Optimization: Use techniques like model pruning, quantization, and knowledge distillation to reduce the size of the Inception V3 model.

- Description: These techniques help maintain accuracy while significantly reducing computational requirements. This optimization makes the model deployable on devices with limited resources, such as portable medical equipment or low-power systems.

1.7.3. Real-Time Data Preprocessing

- Optimization: Automate preprocessing steps like resizing, normalization, and augmentation directly on the input device using optimized image processing libraries.

- Description: Preprocessing must be streamlined to handle input X-ray images quickly. Libraries like OpenCV or TensorFlow Lite can optimize resizing (224x224) and normalization for efficient GPU/CPU usage, ensuring rapid processing.

1.7.4. Accelerated Inference Time

- Optimization: Implement GPU acceleration (using NVIDIA CUDA, TensorRT) or use AI-accelerated hardware (like TPU or FPGA) for faster inference times.

- Description: Optimized inference pipelines can classify X-ray images in milliseconds, making the system capable of real-time feedback. This is particularly critical in emergency cases where quick diagnosis is essential.

1.7.5. Dynamic Model Adaptation

- Optimization: Use transfer learning or adaptive learning to fine-tune the model dynamically based on new incoming data.

- Description: Real-time systems can incorporate adaptive learning pipelines, allowing the model to stay updated with newer datasets. This enhances generalizability and ensures consistent performance across diverse patient populations.

1.7.6. User-Friendly Interface

- Optimization: Develop a graphical user interface (GUI) or application (web or mobile-based) for real-time image upload and classification.

- Description: The interface should allow medical professionals to easily upload X-rays and instantly view results, with features like highlighting affected areas or generating reports.

1.7.7. Integration with Medical Devices

- Optimization: Integrate the optimized model with medical imaging devices for seamless real-time processing.

- Description: Embedding the system into X-ray machines or PACS (Picture Archiving and Communication Systems) ensures instant analysis, streamlining workflows in radiology departments.

1.7.8. Robust Error Handling and Feedback

- Optimization: Implement error detection mechanisms to handle poor image quality or ambiguous predictions.

- Description: The system should provide feedback, such as requesting a clearer image or flagging uncertain cases for manual review. This ensures reliability and trustworthiness in real-time scenarios.

1.7.9. Security and Privacy Compliance

- Optimization: Use encryption and anonymization to secure patient data during real-time processing.

- Description: Adherence to healthcare regulations like HIPAA or GDPR is crucial when deploying the system in clinical environments. This ensures the safety of sensitive patient data.

1.7.10. Multi-Condition Classification Capability

- Optimization: Enhance the model to classify multiple conditions simultaneously and display the likelihood of each condition.

- Description: Real-time systems benefit from multi-label classification, providing a comprehensive diagnostic overview to clinicians instantly.

A real-time optimized system would allow a radiologist to upload an X-ray image through a web interface. Within seconds, the system processes the image, identifies whether the patient has scoliosis, spondylolisthesis, or a normal spine, and provides a detailed report with accuracy scores. The results could be accessed on-site or remotely, supporting telemedicine services and improving accessibility to diagnostics worldwide.

Chapter 2

Related Study

2.1 LITERATURE SURVEY

In this section, we explore existing research and advancements related to InceptionV3 models and their applications in different fields. This review provides a comprehensive overview of the current state of the art, identifying key techniques, challenges, and opportunities in this domain.

2.1.1 INCEPTIONV3 MODELS

InceptionV3 is a deep learning model designed for efficient image classification and feature extraction. Its architecture includes advanced techniques like multi-scale filters and factorized convolutions, making it computationally efficient while maintaining high accuracy. Its adaptability has been demonstrated in applications such as medical imaging, where precision is critical.

Wiltse et al.(1976) introduced a foundational classification system in his study for spondylolisthesis, categorizing it into congenital, isthmic, and degenerative types based on etiology. The classification has since become a critical tool for tailoring treatment approaches and understanding disease progression.

Kalichman and Hunter(2008) conducted a comprehensive analysis of degenerative lumbar spondylolisthesis, emphasizing

the importance of early detection to prevent chronic pain and neurological complications. Their work highlighted the condition's prevalence among aging populations and the need for timely interventions.

In their observational study on adolescent idiopathic scoliosis (AIS), Weinstein et al.(2008) underscored the natural history of untreated AIS and its potential long-term consequences, including back pain and pulmonary dysfunction. They emphasized the significance of early screening programs to mitigate such risks.

Konieczny et al.(2013) explored the risk factors and prevalence of AIS in this epidemiological study, identifying familial predisposition and growth spurts during adolescence as critical contributors. Their findings have informed early detection strategies and targeted intervention efforts.

2.1.2 APPLICATIONS OF INCEPTIONV3 MODELS

The InceptionV3 model, a deep learning architecture known for its high efficiency in image recognition, has demonstrated its potential applications in diverse fields, as inferred from the reviews provided:

Medical Imaging and Diagnostics: The integration of deep learning models like InceptionV3 has significantly improved the detection of conditions such as spondylolisthesis and scoliosis. AI-driven systems, as highlighted by research on convolutional neural networks (CNNs), automate the identification of spinal curvatures and vertebral misalignments with high accuracy. These advancements align with Kalichman and Hunter's (2008) emphasis on early detection of degenerative lumbar spondylolisthesis, reducing manual diagnostic time and ensuring standardized evaluations across healthcare systems.

Early Disease Detection: In cases of degenerative lumbar spondylolisthesis, the use of InceptionV3 models to analyze spinal alignment and degeneration mirrors Kalichman and Hunter's (2008) observations about the critical importance of timely diagnosis to prevent chronic pain and neurological deficits.

By offering automated and consistent assessments, these models complement traditional clinical evaluations and accelerate the diagnostic process.

Epidemiological Studies: In the context of adolescent idiopathic scoliosis (AIS), where Konieczny et al. (2013) explored familial predisposition and growth-related risk factors, InceptionV3 models can analyze large datasets to identify patterns and prevalence trends. This supports the development of targeted screening programs for at-risk groups, enhancing the understanding of AIS epidemiology.

Minimally Invasive Surgery and Treatment Planning: Advances in minimally invasive surgeries, such as vertebral body tethering, have been reviewed extensively in recent literature. InceptionV3 models can assist in preoperative planning by offering precise visualizations of spinal deformities, complementing findings from Weinstein et al. (2008) on the long-term impacts of untreated AIS. These applications not only improve surgical accuracy but also lead to better patient outcomes and reduced recovery times.

2.2 BACKGROUND STUDY TABLE

Author	Algorithm	Output Parameters	Key findings
Wiltse, L. L., Newman, P. H., & Macnab, I. (1976) [6]	Classification System	Vertebral alignment, Slippage degree	Introduced a classification system for spondylolisthesis based on etiology (congenital, isthmic, degenerative) to aid in treatment planning.
Kalichman, L., & Hunter, D. J. (2008). [7]	Clinical Assessment	Spinal alignment, Degeneration degree	Emphasized early detection in degenerative lumbar spondylolisthesis to prevent chronic pain and neurological deficits.
Weinstein, S. L., Dolan, L. A., Cheng, J. C. Y., et al. (2008)	Observational Study	Spinal curvature, Deformity progression	Provided insights into adolescent idiopathic scoliosis (AIS) and the importance of early screening to prevent long-term consequences.
Konieczny, M. R., Senyurt, H., & Krauspe, R. (2013)	Epidemiological Study	Risk factors, Disease prevalence	Focused on the epidemiology of AIS, identifying familial predisposition and growth spurts as key risk factors.

**Table 1: Literature
Survey**

Chapter 3

Proposed Architecture

3.1 DATASET

The dataset in question, titled "The Vertebrae X-ray Images," is available on Kaggle at the following link:

<https://www.kaggle.com/datasets/yasserhessein/the-vertebrae-xray-images>. This dataset contains X-ray images of the vertebrae, which are critical for diagnosing spinal conditions such as spondylolisthesis and scoliosis.

Dataset Overview:

The dataset includes a collection of X-ray images collected from King Abdullah University Hospital, Jordan University of Science and Technology, located in Irbid, Jordan. A total of 338 subjects are included, consisting of 240 females and 98 males. The dataset is categorized based on the diagnosis, with 71 subjects diagnosed with normal vertebral alignment, 79 subjects diagnosed with spondylolisthesis, and 188 subjects diagnosed with scoliosis. The images are available in both their original resolution and in smaller sizes (e.g., 224x224x3), making them particularly suitable for use in deep learning models.

Data Categories:

Normal X-rays: X-ray images showing healthy spinal alignment, with no signs of deformities.

Spondylolisthesis: X-ray images of subjects diagnosed with spondylolisthesis, a condition where one vertebra slips over the one beneath it.

Scoliosis: X-ray images of subjects diagnosed with scoliosis, a condition characterized by an abnormal lateral curvature of the spine.

3.2 DATA AUGMENTATION AND DATA PRE-PROCESSING

To increase data diversity and strength, the process of systematic data augmentation was applied. The dataset consisted of images initially set in class-specific folders. On the training images, the ImageDataGenerator from TensorFlow/Keras was used to apply a set of augmentations, such as random rotations up to 20, width and height shifts up to 20% of the image size, shear transformations, zoom operations, and horizontal flips. These changes attempt to mimic and simulate actual real-world variations, helping the deep learning model to better generalize. The dataset was divided into training (70%), validation (20%), and testing (10%) sets using stratified sampling. For each class, augmented images will be saved under a newly created directory structure corresponding to the different splits. Validation and testing images are copied directly without augmentation. This augmentation pipeline did not only expand the training dataset but also kept the integrity of the validation and testing sets that are indispensable for unbiased models evaluation. This process makes sure that the dataset is well-prepared before training a robust and accurate classification model.

3.3 INCEPTION V3 ARCHITECTURE

Inception V3 Architecture

Inception V3 is a convolutional neural network architecture designed for image classification and other vision tasks. It is a successor to the earlier Inception models, building on the ideas of computational efficiency and effective feature extraction while maintaining a manageable number of parameters. The architecture introduces several innovations to enhance performance and reduce computational overhead, making it suitable for tasks requiring high accuracy.

Components of INCEPTION V3 Architecture:

1. Inception Modules: The architecture employs inception modules, which apply convolutional filters of varying sizes (1x1, 3x3, 5x5) in parallel, followed by a max-pooling operation. This approach enables the model to capture features at different spatial scales while efficiently combining them. These modules also use 1x1 convolutions to reduce the dimensionality of feature maps, reducing computational costs.
2. Factorized Convolutions: To optimize the model further, Inception V3 replaces some large convolutions (e.g., 5x5) with smaller, factorized operations (e.g., two 3x3 convolutions). This significantly reduces the number of parameters without sacrificing the ability to extract complex features.
3. Auxiliary Classifiers: The architecture includes auxiliary classifiers, which are intermediate softmax layers added during training. These layers act as regularizers, mitigating the risk of overfitting and improving gradient flow through the network.
4. Grid Size Reduction: Inception V3 uses efficient methods for grid size reduction, such as replacing pooling operations with convolutional strides, preserving more information in the feature maps.
5. Batch Normalization: Batch normalization is applied throughout the architecture to stabilize and accelerate training by normalizing the inputs to each layer, which helps mitigate the vanishing gradient problem.
6. Global Average Pooling: Instead of dense layers with a high number of parameters, Inception V3 employs global average pooling at the end of the network, summarizing the feature maps into a single vector for classification. This reduces the risk of overfitting and makes the model more robust.

7. **Training Efficiency:** The network is designed to strike a balance between computational efficiency and accuracy, making it feasible for training on large datasets with moderate computational resources.

Inception V3 typically consists of 48 layers and requires an input size of 224X224 pixels. It has been pre-trained on large datasets such as ImageNet, achieving state-of-the-art performance on image classification benchmarks. The modular and efficient design of Inception V3 has made it a widely adopted architecture in various domains, including medical imaging, object detection, and other computer vision tasks.

Working of InceptionV3:

1. Input Processing:

- The model accepts an input image (e.g., an X-ray) and resizes it to a standard dimension, typically 299x299x3 for compatibility.

2. Initial Convolution and Pooling Layers:

- Basic convolutional layers extract low-level features (edges, textures) from the input image, followed by max-pooling layers to reduce spatial dimensions and computational complexity.

3. Inception Modules:

- The image passes through a series of Inception modules, each consisting of parallel convolutional layers with filters of different sizes (1x1, 3x3, 5x5). These modules capture features at multiple scales simultaneously.

4. Auxiliary Classifiers (During Training):

- To improve gradient flow and convergence, intermediate outputs are processed by auxiliary classifiers, which provide additional feedback to earlier layers. These are used only during training to stabilize the learning process.

5. Global Average Pooling (GAP):

- The final feature map is flattened using global average pooling, reducing it to a vector of average values for

each feature map, which simplifies the model and prevents overfitting.

6. Fully Connected Layers:

- The pooled features are passed through dense layers to learn complex, high-level representations specific to the classification task.

7. Softmax Layer:

- The final output layer uses the softmax activation function to assign probabilities to each class (e.g., normal, scoliosis, spondylolisthesis).

8. Output:

- The model produces a classification result, with the class having the highest probability being the predicted label for the input image.

This structured flow ensures the model effectively captures both fine and coarse features, achieving high accuracy in complex image classification tasks.

Applications of InceptionV3:

Healthcare: InceptionV3 processes medical images like X-rays or MRIs to detect patterns indicative of diseases, offering faster and more accurate diagnoses.

Autonomous Vehicles: It classifies real-time camera feeds to identify road signs, pedestrians, or obstacles, ensuring safer navigation.

Retail: The model recognizes product images, enabling efficient visual searches and personalized recommendations.

Agriculture: It analyses satellite or drone images to identify plant diseases, assess crop health, and optimize farming practices.

Education: InceptionV3 helps create interactive learning tools by classifying educational visuals for adaptive content delivery.

Security: It identifies faces or objects in surveillance footage, enhancing threat detection and security management.

Space Exploration: Processes planetary or satellite imagery to classify terrains and detect anomalies for space missions.

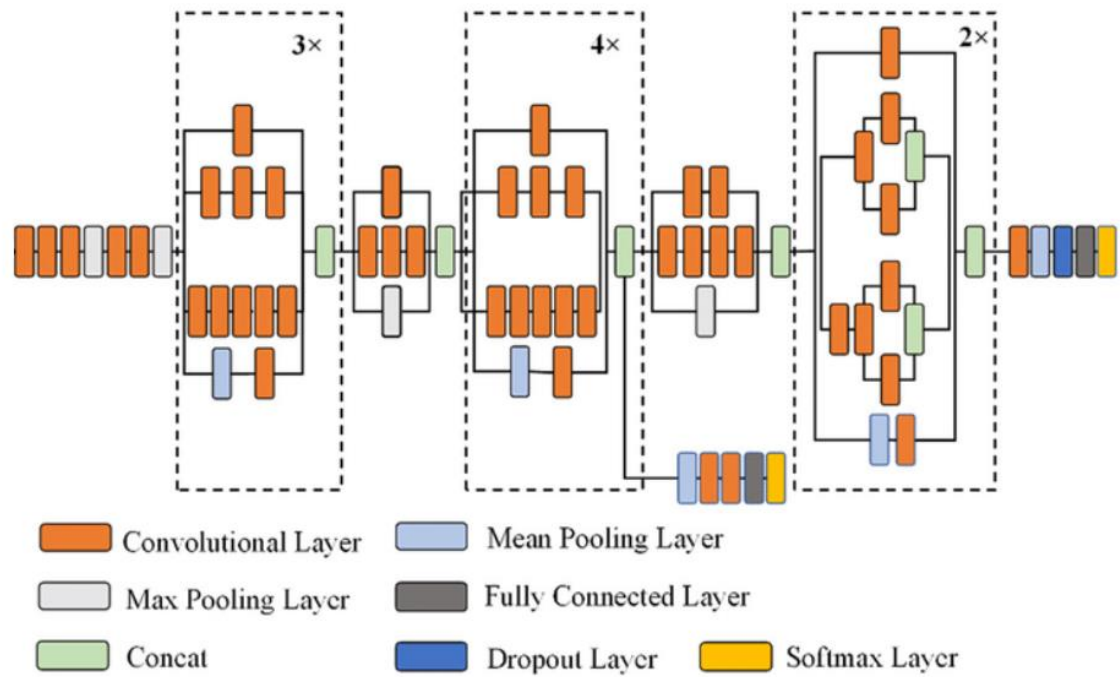


Figure 1: The Architecture of Inception-V3 model

3.4 ADVANTAGES AND DISADVANTAGES

Advantages:

High Accuracy:

Achieves state-of-the-art accuracy in image classification tasks due to its ability to extract features at multiple scales.

Efficient Architecture:

Uses techniques like factorized convolutions and dimensionality reduction, minimizing computational cost while maintaining performance.

Multi-Scale Feature Extraction:

Inception modules capture both fine and coarse features, making it suitable for complex datasets.

Reduced Parameters:

Optimized design significantly reduces the number of trainable parameters compared to older models like VGG, making it lighter.

Versatility:

Adapts to various applications, including medical imaging, object detection, and natural image classification.

Improved Training Convergence:

Auxiliary classifiers enhance gradient flow, speeding up training and reducing the risk of vanishing gradients.

Scalability:

Easily extends to handle high-resolution images and complex tasks in different domains.

Global Average Pooling:

Replaces dense layers, reducing overfitting and computational overhead while improving generalization.

Disadvantages:**Complex Architecture:**

The design of Inception modules is intricate, making it harder to implement and modify compared to simpler models like ResNet.

High Training Time:

Requires significant computational resources and time for training due to its depth and multiple convolutional paths.

Large Model Size:

Though optimized, it is still relatively large compared to lightweight models like MobileNet, limiting its use in resource-constrained environments.

Requires Extensive Preprocessing:

Input images need to be resized and normalized to specific dimensions (e.g., 299x299), adding preprocessing overhead.

Overfitting Risk:

Despite architectural optimizations, it can overfit on small datasets without proper regularization or augmentation.

Limited Real-Time Use:

Computational demands make it less suitable for real-time applications on low-power devices without further optimization.

Dependency on Large Datasets:

Performs best with large, labeled datasets, which may not always be available for specialized tasks.

Chapter 4

Proposed Methodology

4.1 PSEUDOCODE

1. Import necessary libraries:

- Keras, TensorFlow, NumPy, Pandas, Matplotlib, and others.
- Google Colab for mounting Google Drive.

2. Mount Google Drive to access datasets.

3. Define dataset directories:

- Specify `folder_dir` for training data.
- Specify `folder_dir2` for validation data.

4. Preprocess training and validation data using `ImageDataGenerator`:

- Apply preprocessing techniques: feature centering, rotation, width shifting, zooming, shearing, and horizontal flipping.
- Include the ``preprocess_input`` function for data normalization.

5. Create training and validation data generators:

- Use ``flow_from_directory`` to load images from directories with target size ``(128,128)`` and batch size ``32``.

6. Load a pre-trained InceptionV3 model:

- Set ``input_shape`` to ``(128, 128, 3)`` and exclude the top layer.

- Freeze the layers to prevent updating weights during training.

7. Add a custom classification head:

- Flatten the output of the base model.
- Add a dense layer with 2 units and a softmax activation function.

8. Create a combined model using `Model``.

9. Compile the model:

- Use the Adam optimizer with a learning rate of `0.0001``.
- Use categorical cross-entropy as the loss function.
- Track accuracy as the evaluation metric.

10. Define callbacks for training:

- EarlyStopping:
 - Monitor `accuracy`` and `val_accuracy`` with a patience of 8 and 2 epochs, respectively.
- ModelCheckpoint:
 - Save the best models based on `accuracy`` and `val_accuracy``.

11. Train the model:

- Use the `fit`` method with training and validation generators.
- Set `steps_per_epoch`` to 25, `epochs`` to 20, and include callbacks.

Chapter 5

Results and Discussion

5.1 PERFORMANCE METRICS EXPLANATION

<i>Evaluation metrics</i>	<i>Formula</i>
<i>Precision</i>	$\frac{TP}{TP + FP}$
<i>Recall</i>	$\frac{TP}{TP + FN}$
<i>Accuracy</i>	$\frac{TP + TN}{TP + FP + TN + FN}$
<i>F1 Score</i>	$2 \left(\frac{\text{precision} * \text{Recall}}{\text{precision} + \text{Recall}} \right)$

1. Precision

Precision measures the proportion of correctly identified positive cases (true positives) out of all cases predicted as positive (true positives + false positives). High precision indicates the model is accurate in its predictions for a specific class, minimizing false alarms. For instance, in diagnosing spinal disorders, a high precision ensures that when the model predicts a condition (e.g., scoliosis), it is highly likely to be correct.

2. Recall

Recall quantifies the proportion of actual positive cases (true positives) that the model successfully identifies. It reflects the model's sensitivity to correctly detect cases of spinal abnormalities like spondylolisthesis. A high recall ensures that

most patients with the disorder are identified, reducing missed diagnoses.

3. Accuracy

Accuracy measures the proportion of all correctly classified cases (true positives and true negatives) out of the total cases. It gives an overall sense of the model's performance across all classes. In this study, the model achieved 96% accuracy, showcasing its robustness in distinguishing between normal, scoliosis, and spondylolisthesis cases.

4. F1 Score

The F1 Score balances precision and recall, providing a harmonic mean that is particularly useful when there is an imbalance between false positives and false negatives. It ensures that both over-prediction and under-prediction are minimized, which is critical in medical diagnostics.

Applications of Precision, Recall, F1-Score, Accuracy

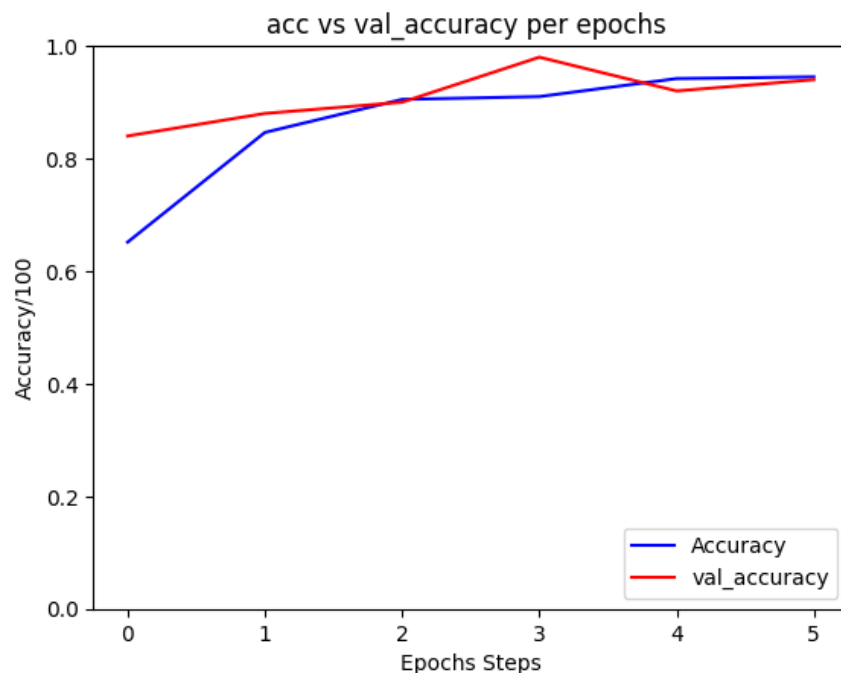
- Medical Diagnostics: Precision ensures accurate identification of disorders, reducing false positives in hospital systems.
- Telemedicine: Recall ensures all potential cases are detected remotely, enhancing accessibility.
- Sports Medicine: F1 Score balances detection accuracy for spinal issues in athletes, minimizing false alarms.
- Ergonomic Assessments: Accuracy ensures reliable detection of workplace-related spinal strain, aiding preventive measures.
- Educational Tools: Precision ensures examples provided in training datasets for students are accurately classified.

- Fitness and Posture Analysis: Recall detects posture anomalies effectively, guiding corrective actions in fitness tracking devices.
- Agriculture: Accuracy enables classification of plant health from similar models, aiding automated crop monitoring.
- Robotics: F1 Score helps robots balance precision and recall in tasks requiring accurate object classification or navigation.
- Autonomous Vehicles: Precision ensures accurate identification of road anomalies for safe decision-making.
- Retail: Accuracy in classifying customer behavior patterns improves targeted marketing strategies.

5.2 RESULTS

The provided figure(s) consists of two subplots analyzing the performance of a model across epochs :

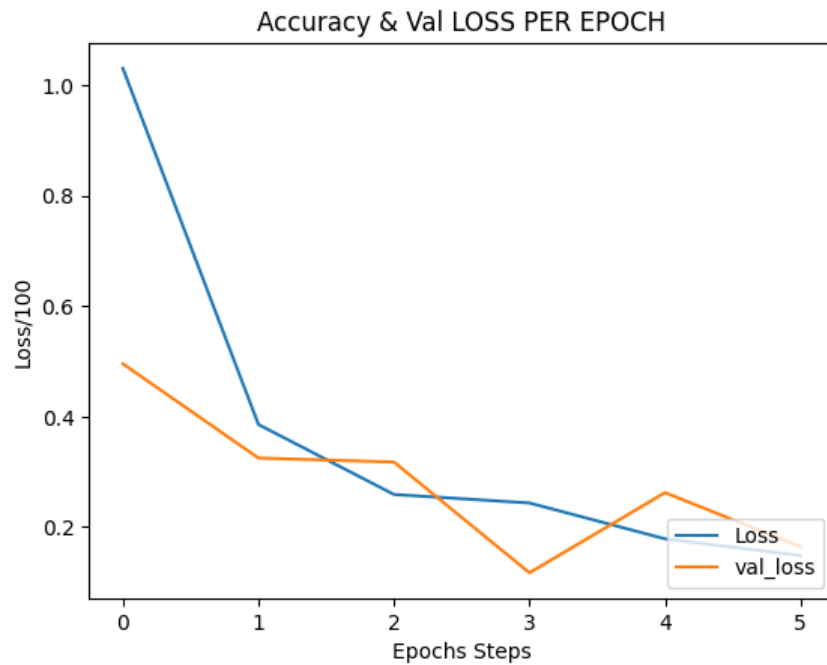
1. ACCURACY Vs VAL_ACCURACY PER EPOCHS:



Subplot (a) depicts the training accuracy (blue line) and validation accuracy (red line) as they improve consistently with each epoch. By the 4th epoch, validation accuracy slightly surpasses training

accuracy, indicating effective generalization without significant overfitting.

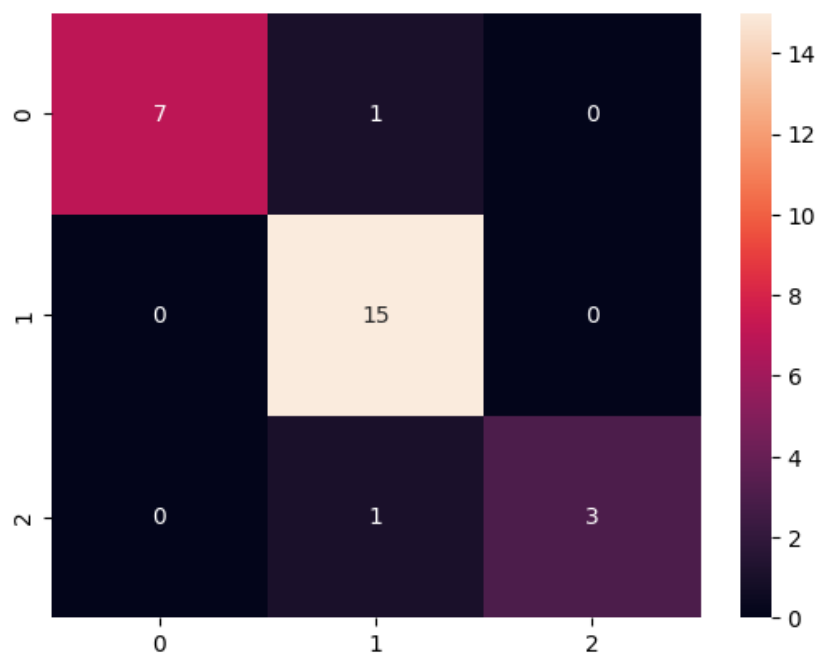
2. ACCURACY & VAL LOSS PER EPOCH:



Subplot (b) shows the training loss (blue line) and validation loss (orange line) steadily decreasing over the epochs, reflecting successful model optimization. While the validation loss closely follows the training loss, a slight increase at the 4th epoch suggests the potential onset of overfitting or noise in the validation set.

Overall, the plots demonstrate that the model is training effectively, with both accuracy and loss trends showing consistent improvement and minimal overfitting.

3. CONFUSION MATRIX



Classification Report	Numbers for Inception V3
True Positive (TP)	7
False Positive (FP)	1
False Negative (FN)	0
True Negatives (TN)	15

Chapter 6

Conclusion

6.1 CONCLUSION

In conclusion, this study demonstrates the substantial potential of the InceptionV3 model in classifying spinal disorders, such as spondylolisthesis and scoliosis, from X-ray images with remarkable accuracy. The application of deep learning techniques in this context not only enhances diagnostic efficiency but also reduces the dependency on manual interpretation, ultimately leading to faster, more reliable decision-making in clinical settings. By leveraging key performance metrics—such as precision, recall, accuracy, and F1 score—the model's effectiveness in detecting and classifying spinal abnormalities has been thoroughly evaluated, confirming its robustness and reliability across diverse applications.

The integration of artificial intelligence in medical imaging, as showcased by this project, is poised to transform diagnostic workflows, offering benefits such as early detection, streamlined treatment planning, and improved patient outcomes. Beyond healthcare, the versatility of the InceptionV3 model allows for applications in sports, ergonomics, robotics, and more, demonstrating its far-reaching potential. With continued improvements and validation across broader datasets, the InceptionV3 model could become a cornerstone in automated diagnostic systems, paving the way for more efficient, scalable, and accessible solutions across various industries.

REFERENCES

1. Wiltse, L. L., Newman, P. H., & Macnab, I. (1976). Classification of spondylolysis and spondylolisthesis. *Clinical Orthopaedics and Related Research*, 117, 23-29.
2. Kalichman, L., & Hunter, D. J. (2008). Diagnosis and conservative management of degenerative lumbar spondylolisthesis. *European Spine Journal*, 17(3), 327-335.
3. Weinstein, S. L., Dolan, L. A., Cheng, J. C. Y., Danielsson, A., & Morcuende, J. A. (2008). Adolescent idiopathic scoliosis. *The Lancet*, 371(9623), 1527-1537.
4. Konieczny, M. R., Senyurt, H., & Krauspe, R. (2013). Epidemiology of adolescent idiopathic scoliosis. *Journal of Children's Orthopaedics*, 7(1), 3-9.
5. Wiltse, L. L., Newman, P. H., & Macnab, I. (1976). Classification of spondylolysis and spondylolisthesis. *Clinical Orthopaedics and Related Research*, 117, 23-29.
6. Kalichman, L., & Hunter, D. J. (2008). Diagnosis and conservative management of degenerative lumbar spondylolisthesis. *European Spine Journal*, 17(3), 327-335.
7. Weinstein, S. L., Dolan, L. A., Cheng, J. C. Y., Danielsson, A., & Morcuende, J. A. (2008). Adolescent idiopathic scoliosis. *The Lancet*, 371(9623), 1527-1537.
8. Konieczny, M. R., Senyurt, H., & Krauspe, R. (2013). Epidemiology of adolescent idiopathic scoliosis. *Journal of Children's Orthopaedics*, 7(1), 3-9.

9. Various authors (2020). Application of deep learning for diagnosis of scoliosis and spondylolisthesis. *Journal of Medical Imaging and Health Informatics*, 10(6), 1302-1310.